

Macguire McDuff

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EDUCATION

University of Florida

August 2021 – May 2025

Bachelor of Science in Computer Science, Herbert Wertheim College of Engineering

Gainesville, FL

- GPA: 3.4; Dean's List (multiple semesters) and SEC Athletic Honor Roll (all semesters)
- 5x NCAA Champion, 12x NCAA All-American, and Current NCAA Record Holder

PROFESSIONAL EXPERIENCE

Associate Software Developer

June 2025 – Present

iPipeline

Fort Lauderdale, Florida

- Architected an asynchronous prompt execution service using Python, AWS DynamoDB, and Step Functions, exposing **process** and **status** REST endpoints to decouple LLM requests from response handling and manage state across long-running AI workflows.
- Built and containerized a Utilities API platform using Python, FastAPI, and AWS Fargate (ECR), standardizing internal service patterns; included a Jira integration service with secure credential handling via AWS Secrets Manager.
- Queried AWS Athena and Lake Formation data lakes to extract and prepare structured datasets for AI model ingestion, maintaining critical data pipelines through an AWS outage in collaboration with the internal data team.
- Led end-to-end development of a Salesforce Support Agent MCP with asynchronous tooling and Step Functions integrations for Co-Pilot Studio compatibility, improving support team workflows.
- Built an AWS Bedrock token vending CLI tool using Python, Lambda, and Terraform, enabling time-scoped developer access to Claude AI with cost governance via AWS Cost Explorer.

Software Engineer Intern

May 2023 – July 2023

Advantus Corp

Jacksonville, FL

- Enhanced and refactored proprietary PHP-based applications within an Agile/Scrum team, implementing performance optimizations for a RESTful API.
- Designed complex SQL queries, stored procedures, triggers, and PL/SQL packages with schema and performance improvements across Oracle database platforms.

PROJECTS

Sports Betting Intelligence Platform | *Python, MCP, The Odds API, ESPN API, FastMCP*

[Demo](#)

- Built a multi-source data aggregation system integrating The Odds API and ESPN's real-time scoreboard API, consolidating live odds, scores, and team stats across 10+ leagues into a unified query interface.
- Deployed a custom MCP server exposing 8 structured tools for odds retrieval, arbitrage detection, market efficiency analysis, and real-time game state tracking via Claude AI.
- Implemented a bookmaker margin analysis pipeline parsing American-format moneylines across sportsbooks to surface +EV betting opportunities with cross-book probability differentials.

ML Student Performance Predictor | *Python, Pandas, NumPy, scikit-learn, TensorFlow*

[GitHub](#)

- Led a team of four to build regression and classification models predicting student exam outcomes across 6,607 records, achieving 97.03% accuracy using scikit-learn and TensorFlow.
- Analyzed 20 performance factors and applied preprocessing pipelines; evaluated models with MSE, R^2 , accuracy, and F1 score metrics.

OTHER EXPERIENCE

USA National Team Member

April 2024 – July 2024

United States Olympic & Paralympic Committee

Colorado Springs, CO

SKILLS

Languages: Python, C++, Java, JavaScript, HTML/CSS

Frameworks & Tools: FastAPI, React, Git, Docker, Node.js, Angular, TensorFlow, scikit-learn, Linux, SQL, Postman, MCP

Cloud & DevOps: AWS (Lambda, S3, Bedrock, CloudWatch, DynamoDB, Step Functions, Fargate, ECR, Athena, Lake Formation, Secrets Manager), Terraform, Terragrunt